Chap 1. Basic Concepts of Probability Theory

Outline

- Random experiments and Probability - Sec. 1.1, 1.2
- The axioms of probability - Sec. 1.3, 1.4
- Conditional probability - Sec. 1.5
- Independence of events - Sec. 1.6
- Sequential experiments - Sec. 1.7, 1.8, 1.9
- Applications

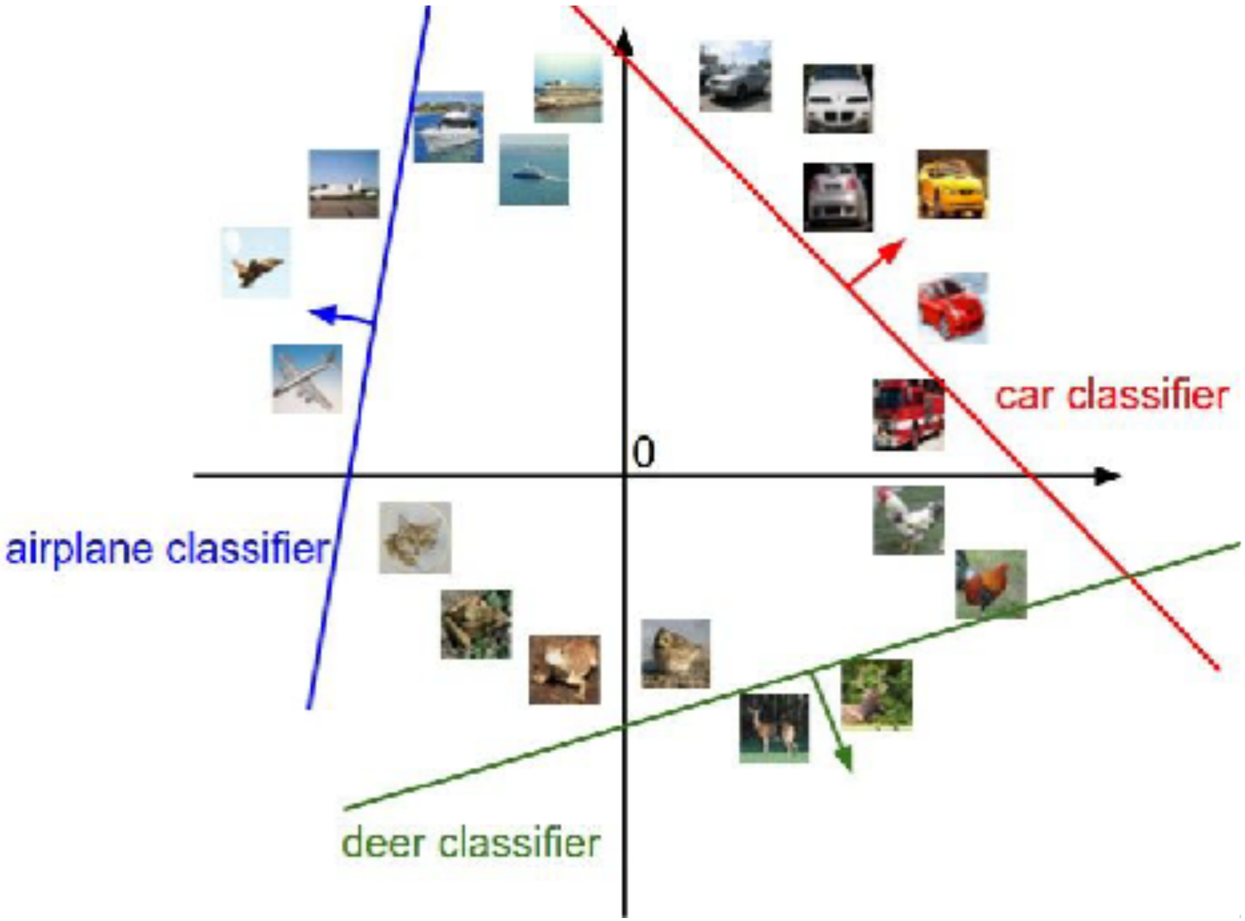
Random experiments, Set Theory and Probability

Sections 1.1 and 1.2

What is Probability

- Describe phenomena that cannot be described with certainty because of the complexity of the underlying physical process.
- Different from your study of the deterministic sciences, e.g., the laws of classical mechanics.
- A number between 0 and 1.
- Probabilities are assigned based on observations, sometimes experience.
- Mathematical basis is in the theory of sets.

Image Recognition



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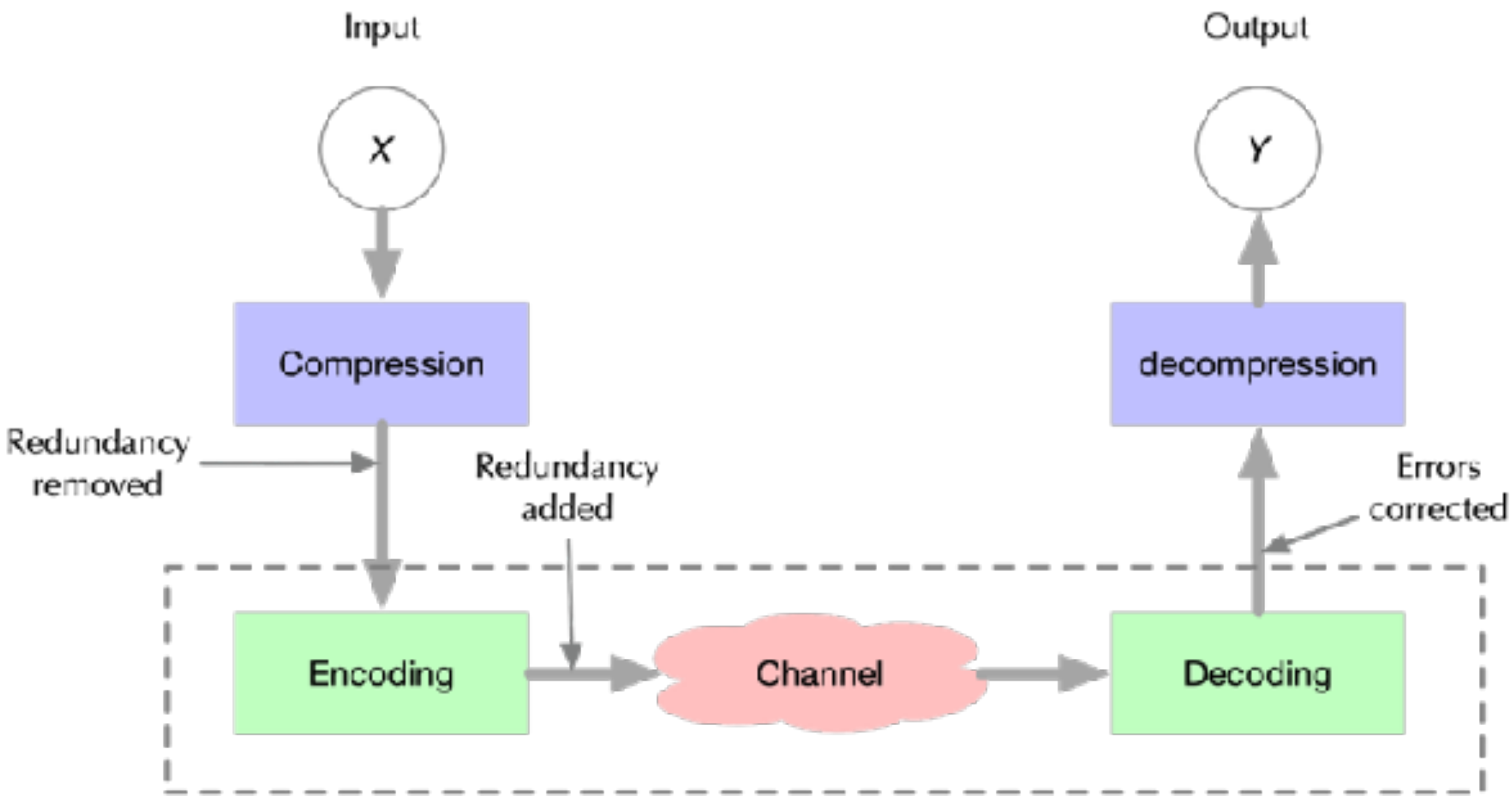
Communication Channel



Credit: https://www.entropy.energy/scholar/node/communication

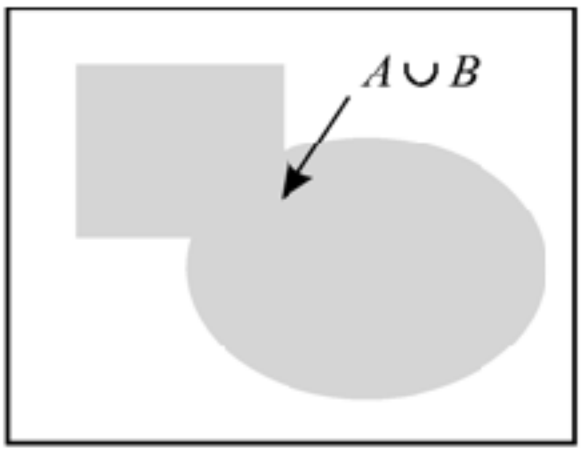
Section 1.1

Set Theory

1.1 Comment: Sets

A *set* is a collection of things. We use capital letters to denote sets. The things that together make up the set are *elements*. When we use mathematical notation to refer to set elements, we usually use small letters. Thus we can have a set A with elements x, y, and z. The symbol $\in$ denotes set inclusion. Thus $x \in A$ means "x is an element of set A." The symbol $\notin$ is the opposite of $\in$. Thus $c \notin A$ means "c is not an element of set A."

1.1 Comment: Set Union

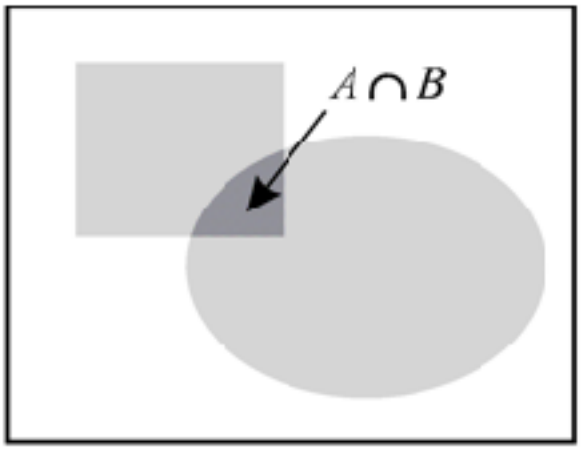


The *union* of sets A and B is the set of all elements that are either in A or in B, or in both. The union of A and B is denoted by $A \cup B$. In this Venn diagram, $A \cup B$ is the complete shaded area. Formally, the definition states

$$x \in A \cup B \text{ if and only if } x \in A \text{ or } x \in B.$$

The set operation union corresponds to the logical "or" operation.

1.1 Comment: Set Intersection

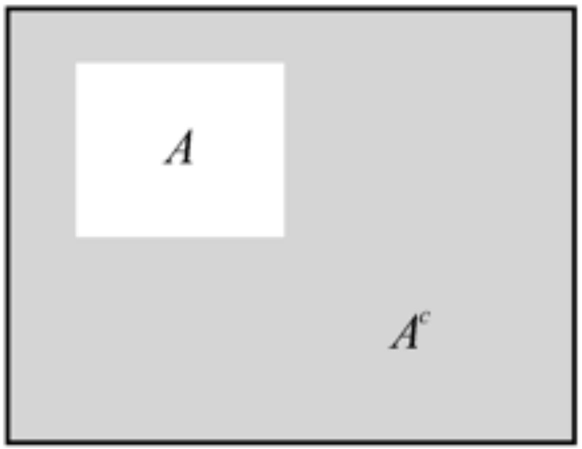


The *intersection* of two sets A and B is the set of all elements which are contained both in A and B. The intersection is denoted by $A \cap B$. Another notation for intersection is AB. Formally, the definition is

$$x \in A \cap B \text{ if and only if } x \in A \text{ and } x \in B.$$

The set operation intersection corresponds to the logical "and" function.

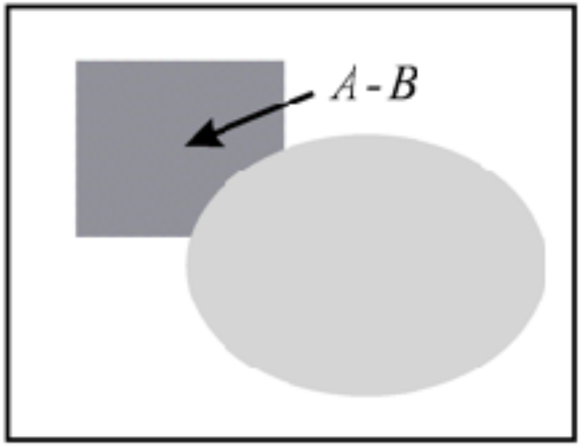
1.1 Comment: Set Complement



The *complement* of a set A, denoted by A^c, is the set of all elements in S that are not in A. The complement of S is the null set ϕ. Formally,

$$x \in A^c \text{ if and only if } x \notin A.$$

1.1 Comment: Set Difference

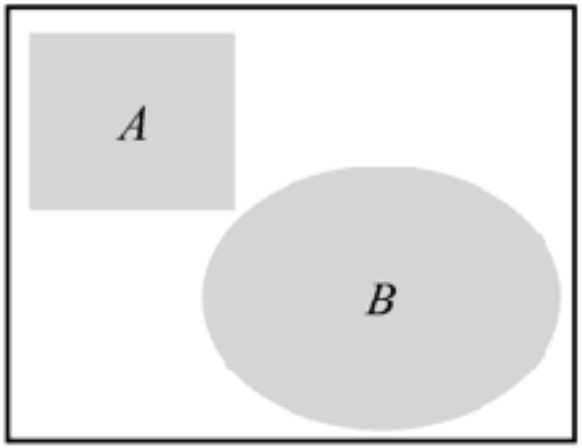


A fourth set operation is called the *difference*. It is a combination of intersection and complement. The *difference* between A and B is a set $A - B$ that contains all elements of A that are *not* elements of B. Formally,

$$x \in A - B \text{ if and only if } x \in A \text{ and } x \notin B$$

Note that $A - B = A \cap B^c$ and $A^c = S - A$.

1.1 Comment: Mutually Exclusive Sets

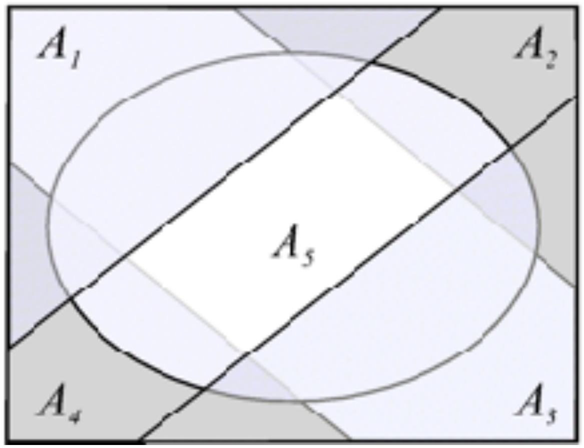


A collection of sets $A_1, \ldots, A_n$ is *mutually exclusive* if and only if

$$A_i \cap A_j = \phi, \qquad i \neq j. \tag{1.8}$$

When there are only two sets in the collection, we say that these sets are *disjoint*. Formally, A and B are disjoint if and only if $A \cap B = \phi$.

1.1 Comment: Collectively Exhaustive Sets



A collection of sets $A_1, \ldots, A_n$ is *collectively exhaustive* if and only if

$$A_1 \cup A_2 \cup \cdots \cup A_n = S. \qquad (1.9)$$

Theorem 1.1

De Morgan's law relates all three basic operations:

$$(A \cup B)^c = A^c \cap B^c.$$

Section 1.2

Applying Set Theory to Probability

1.2 Comment: Experiments

An experiment consists of a *procedure* and *observations*. There is uncertainty in what will be observed; otherwise, performing the experiment would be unnecessary. Some examples of experiments include

1. Flip a coin. Did it land with heads or tails facing up?
2. Walk to a bus stop. How long do you wait for the arrival of a bus?
3. Give a lecture. How many students are seated in the fourth row?
4. Transmit one of a collection of waveforms over a channel. What waveform arrives at the receiver?
5. Transmit one of a collection of waveforms over a channel. Which waveform does the receiver identify as the transmitted waveform?

Example 1.1

An experiment consists of the following procedure, observation, and model:

- Procedure: Flip a coin and let it land on a table.
- Observation: Observe which side (head or tail) faces you after the coin lands.
- Model: Heads and tails are equally likely. The result of each flip is unrelated to the results of previous flips.

Example 1.2

Flip a coin three times. Observe the sequence of heads and tails.

Example 1.3

Flip a coin three times. Observe the number of heads.

Definition 1.1 Outcome

An outcome *of an experiment is any possible observation of that experiment.*

Definition 1.2 Sample Space

The sample space *of an experiment is the finest-grain, mutually exclusive, collectively exhaustive set of all possible outcomes.*

Example 1.4

- The sample space in Example 1.1 is $S = \{h, t\}$ where h is the outcome "observe head," and t is the outcome "observe tail."

- The sample space in Example 1.2 is

$$S = \{hhh, hht, hth, htt, thh, tht, tth, ttt\} \tag{1.13}$$

- The sample space in Example 1.3 is $S = \{0, 1, 2, 3\}$.

Definition 1.3 Event

An event *is a set of outcomes of an experiment.*

Example 1.6

Suppose we roll a six-sided die and observe the number of dots on the side facing upwards. We can label these outcomes $i = 1, \ldots, 6$ where i denotes the outcome that i dots appear on the up face. The sample space is $S = \{1, 2, \ldots, 6\}$. Each subset of S is an event. Examples of events are

- The event $E_1 = \{\text{Roll 4 or higher}\} = \{4, 5, 6\}$.
- The event $E_2 = \{\text{The roll is even}\} = \{2, 4, 6\}$.
- $E_3 = \{\text{The roll is the square of an integer}\} = \{1, 4\}$.

The terminology of set theory and probability

Set Algebra	Probability
Set	Event
Universal set	Sample space
Element	Outcome

The axioms of Probability

Sections 1.3 and 1.4

Definition 1.5 Axioms of Probability

A probability measure $P[\cdot]$ *is a function that maps events in the sample space to real numbers such that*

Axiom 1 *For any event* A, $P[A] \geq 0$.

Axiom 2 $P[S] = 1$.

Axiom 3 *For any countable collection* $A_1, A_2, \ldots$ *of mutually exclusive events*

$$P[A_1 \cup A_2 \cup \cdots] = P[A_1] + P[A_2] + \cdots .$$

Theorem 1.3

For mutually exclusive events A_1 and A_2,

$$P[A_1 \cup A_2] = P[A_1] + P[A_2].$$

Theorem 1.4

If $A = A_1 \cup A_2 \cup \cdots \cup A_m$ and $A_i \cap A_j = \phi$ for $i \neq j$, then

$$P[A] = \sum_{i=1}^{m} P[A_i].$$

Theorem 1.5

The probability of an event $B = \{s_1, s_2, \ldots, s_m\}$ is the sum of the probabilities of the outcomes contained in the event:

$$P[B] = \sum_{i=1}^{m} P[\{s_i\}].$$

Theorem 1.6

For an experiment with sample space $S = \{s_1, \ldots, s_n\}$ in which each outcome s_i is equally likely,

$$P[s_i] = 1/n \qquad 1 \le i \le n.$$

Section 1.4

Some Consequences of the Axioms

Theorem 1.7

The probability measure $P[\cdot]$ satisfies

(a) $P[\phi] = 0$.

(b) $P[A^c] = 1 - P[A]$.

(c) For any A and B (not necessarily disjoint),

$$P[A \cup B] = P[A] + P[B] - P[A \cap B].$$

(d) If $A \subset B$, then $P[A] \le P[B]$.

Quiz 1.4

Monitor a phone call. Classify the call as a voice call (V) if someone is speaking, or a data call (D) if the call is carrying a modem or fax signal. Classify the call as long (L) if the call lasts for more than three minutes; otherwise classify the call as brief (B). Based on data collected by the telephone company, we use the following probability model: $P[V] = 0.7$, $P[L] = 0.6$, $P[VL] = 0.35$. Find the following probabilities:

(1) $P[DL]$

(2) $P[D \cup L]$

(3) $P[VB]$

(4) $P[V \cup L]$

(5) $P[V \cup D]$

(6) $P[LB]$

Quiz 1.4 Solution

We can describe this experiment by the event space consisting of the four possible events VB, VL, DB, and DL. We represent these events in the table:

	V	D
L	0.35	?
B	?	?

In a roundabout way, the problem statement tells us how to fill in the table. In particular,

$$P[V] = 0.7 = P[VL] + P[VB] \quad (1)$$

$$P[L] = 0.6 = P[VL] + P[DL] \quad (2)$$

Since $P[VL] = 0.35$, we can conclude that $P[VB] = 0.35$ and that $P[DL] = 0.6 - 0.35 = 0.25$. This allows us to fill in two more table entries:

	V	D
L	0.35	0.25
B	0.35	?

The remaining table entry is filled in by observing that the probabilities must sum to 1. [Continued]

Quiz 1.4 Solution (Continued 2)

This implies $P[DB] = 0.05$ and the complete table is

	V	D
L	0.35	0.25
B	0.35	0.05

Finding the various probabilities is now straightforward:

(1) $P[DL] = 0.25$

(2) $P[D \cup L] = P[VL] + P[DL] + P[DB] = 0.35 + 0.25 + 0.05 = 0.65.$

(3) $P[VB] = 0.35$

(4) $P[V \cup L] = P[V] + P[L] - P[VL] = 0.7 + 0.6 - 0.35 = 0.95$

(5) $P[V \cup D] = P[S] = 1$

(6) $P[LB] = P[LL^c] = 0$